

SUBRAT ACHARYA

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Objective

Software Engineer with extensive experience in developing and enhancing software solutions for manufacturing and health technology. Proven adaptability, reliability, and a strong willingness to learn, ensuring proficiency in any task when given the opportunity. Demonstrated ability to work independently as well as in a team setting.

Education

Bachelor of Science and Engineering in Biomedical Engineering

University of Iowa, Iowa City, IA

Aug 2018 to May 2022

GPA: 4.0/4.0

Professional Experience

Software Engineer at Avantor

Aug 2022 to Present

Barrington, IL

- Architected and delivered the End of Line (EOL) tester software for Masterflex touchscreen pumps, managing the complete development lifecycle from initial design through production release (Python, Go, Vue.js, TypeScript)
- Spearheaded development of the Masterflex Inventory Management System as a serverless application on AWS, utilizing CloudFormation, Cognito, API Gateway, Lambda, S3, and DynamoDB to collect, store, and analyze production and quality metrics
- Drove key contributions to Tangential Flow Filtration (TFF) monitoring and analysis software, encompassing software architecture design, implementation, and production release (C++, Flutter)
- Optimized EOL tester software by enhancing application performance and refining test cases to efficiently identify hardware failures, yielding measurable improvements in product quality and reduced manufacturing defects
- Facilitated the successful launch of twelve new pump models by conducting thorough research and testing of new features to develop comprehensive test suites for EOL validation
- Automated manual system and smoke testing processes using ROBOT Framework and selenium, greatly increasing testing efficiency and reliability
- Engineered the serial communications module for embedded pump software, enabling RS-232 protocol-based pump control and monitoring capabilities
- Successfully transitioned responsibilities from external vendor Planet Innovation to in-house operations, demonstrating technical ownership and generating \$500,000+ in annual cost savings
- Collaborated with cross-functional teams including software engineers, firmware engineers, mechanical engineers, and end-users to drive continuous software improvements and maintain effective project communication

Software Engineering Intern at iotaMotion

Jun 2020 to Feb

2021

Iowa City, IA

- Researched and developed a sophisticated web application utilizing Django and PostgreSQL to store, analyze, and present auditory health data, which played a role in supporting the company's research initiatives and clinical trials.
- Demonstrated strong problem-solving skills by designing and implementing signal processing algorithms, enabling seamless processing and visualization of electrocochleography data
- Worked closely with engineers and engaged potential end-users to gather feedback, leading to improvements that tailored the application to the users' needs
- Began development on a computer vision based mobile application using React Native to facilitate the collection and storage of auditory health data

Leadership Experience

Teaching Assistant – Engineering Fundamentals II: Circuits

Aug 2019 to Aug 2021

- Led 50-minute discussion classes each week to supplement topics covered in lecture and work through problems
- Held office hours for 5 hours each week to assist students with homework and answer course-related questions
- Led review sessions for up to 60 students before midterms and finals to help students prepare and answer questions

Skills

Python, C++, Java, JavaScript, Golang, MATLAB, HTML, CSS, TypeScript, Linux, PostgreSQL, NoSQL, DynamoDB, Django, Flask, Asynchronous Programming, Vue.js, Node.js, React.js, Flutter, Git, Docker, AWS, Serverless Architecture, Full-stack Web Development, OOP, Agile, Artificial Intelligence, Computer Vision, Signal Processing, Image Processing, Embedded Linux, Unit Testing, Integration Testing, Automation Testing